CSE301 PROJECT
Milestone 2 – Group 2
Domain: Weather and Climate Systems
Scribe

Scribe Question 1: Project System and Objective

What is the probabilistic problem being addressed in your project? Clearly state the system objective and identify the primary sources of uncertainty involved.

Answer:

The project is dealing with the prediction of short-term rainfall when there is uncertainty. Since rainfall and the movement of storms cannot be perfectly predicted, they are considered random. The chief problem is to estimate the chance of excessive level of rainfall.

Basically, what is the probability of the rainfall becoming very heavy?

The main objective of the system is to:

- Model the movement of rainfall as random process.
- Measure and quantify uncertainty of rainfall forecasts.
- Predict and estimate chances of rainfall.
- Give probabilistic bounds for flood warnings and risky decisions.

Instead of predicting one fixed rainfall value, the system calculates:

$$P(R \geq k) = \text{Probability that rainfall exceeds a critical threshold,}$$

Where, k = threshold, R = rainfall.

Primary Sources of Uncertainty:

- **Motion Uncertainty** – Estimation of speed and direction of storm might have errors.
- **Rainfall Variability** – Unpredictable and continuous changes in rainfall intensity.
- **Random Fluctuations (Stochastic Noise)** – Over time, random changes may affect rainfall.
- **Extreme Event Uncertainty** – Uncertain to predict whether rainfall will lead to floods.

Overall, the system is all about being able to predict rainfall in a probabilistic manner to help manage the risks such as flooding.

Scribe Question 2: Key Random Variables and Uncertainty Modeling

Identify the key random variables in your project and describe how uncertainty is modeled for each. Clearly state any probabilistic assumptions currently being made.

Answer:

Motion Perturbation: This is the error in estimating the speed and direction of the storm. Since storms sometimes can change direction suddenly, we use a Laplace distribution which does allow for larger errors than a normal distribution. Simple meaning: Storm movement is uncertain, and at times the error can be great.

$$\Delta X \sim \text{Laplace}(0, b)$$

Probability Density Function:

$$f(x) = \frac{1}{2b} \exp\left(-\frac{|x|}{b}\right)$$

Cumulative Distribution Function:

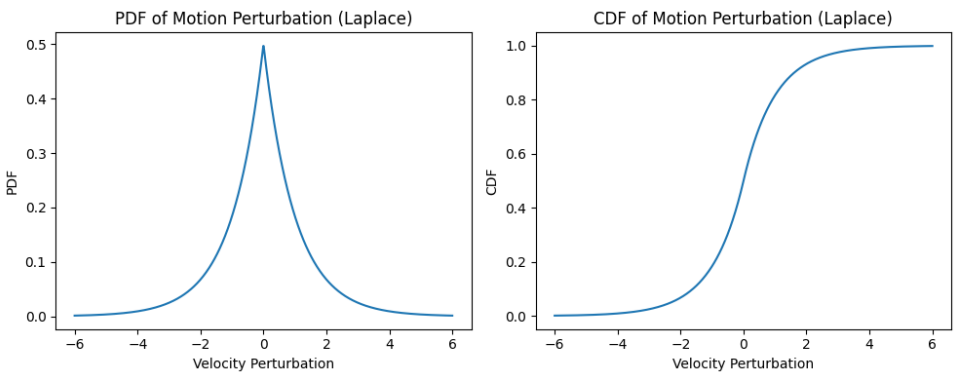
$$F(x) = \begin{cases} \frac{1}{2} \exp\left(\frac{x}{b}\right), & x < 0 \\ 1 - \frac{1}{2} \exp\left(-\frac{x}{b}\right), & x \geq 0 \end{cases}$$

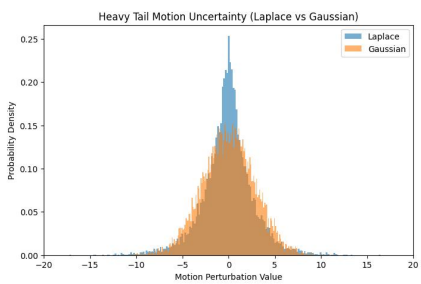
Mean:

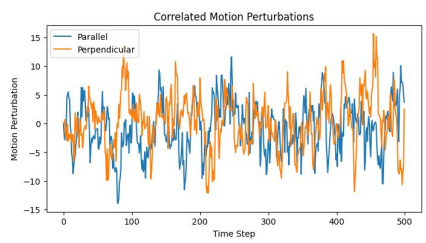
$$\mathbb{E}[\Delta X] = 0$$

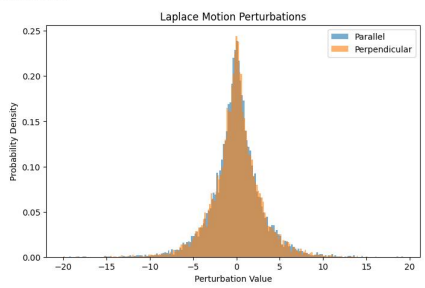
Variance:

$$\text{Var}(\Delta X) = 2b^2$$





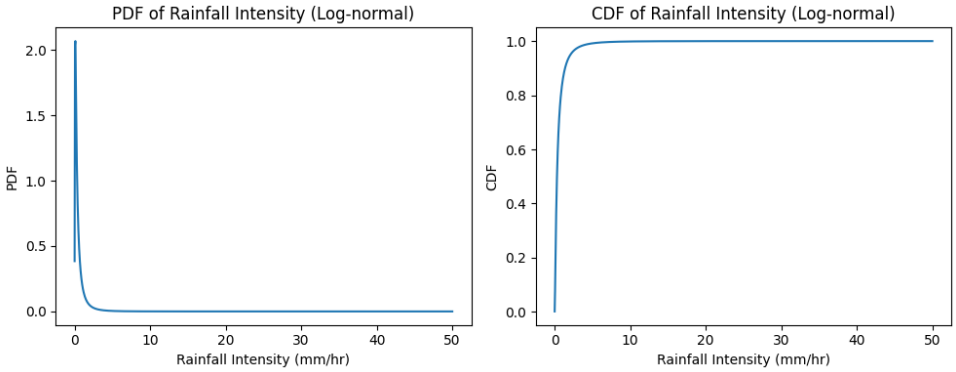


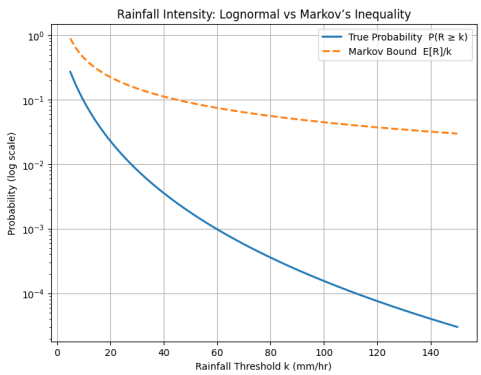


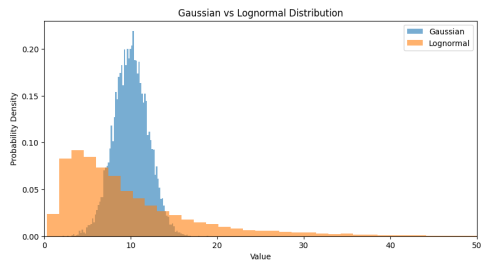
Rainfall Intensity: The amount of rainfall cannot be negative and can be much different. It is modelled through the use of a log-normal distribution due to the potentially large rainfall amount during extreme events. We use Markov's inequality to give an upper bound for: which means the probability that rainfall is greater than a level which is considered dangerous. Simple meaning: Rainfall is always positive and is sometimes very heavy, so we use a model with the capacity for extreme values.

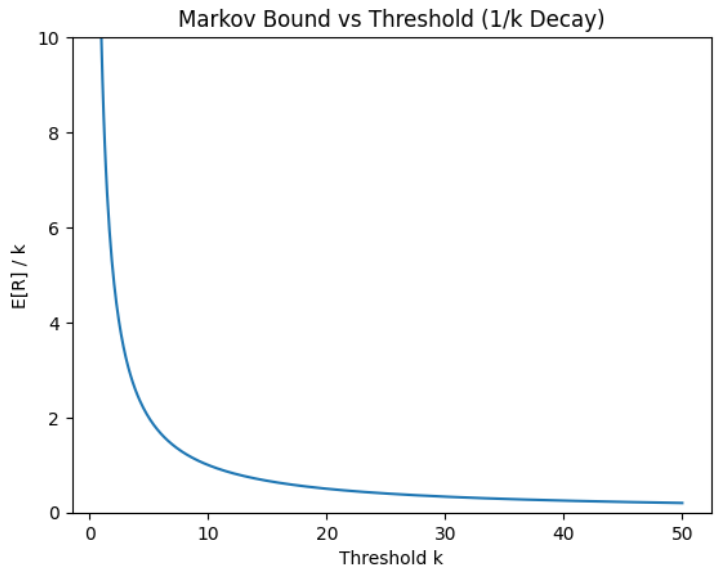
Markov's Inequality:

$$P(R \geq r) \leq \frac{\mathbb{E}[R]}{r}, \quad r > 0$$





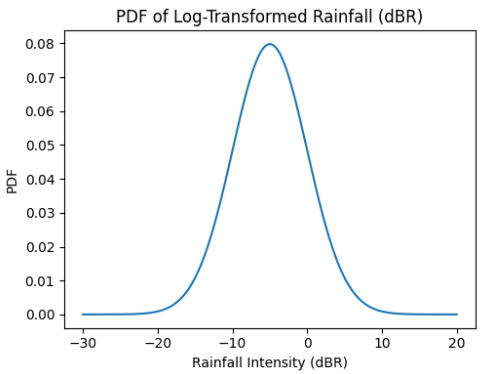


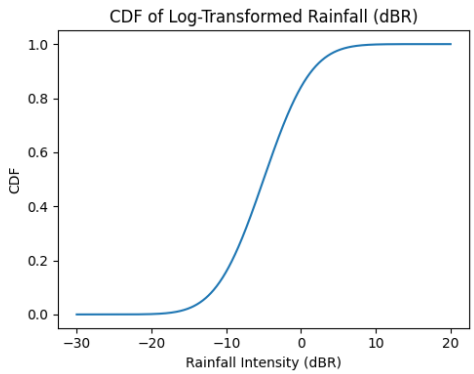


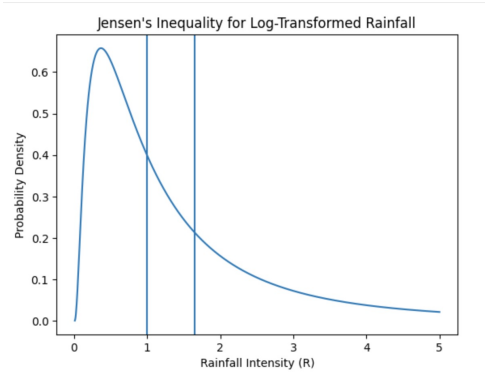
Log-Transformed Rainfall: To make analysis easier we take the logarithm of rainfall. After transformation, it is normally distributed. In order to understand how use of log changes the expected value, we use Jensen's Inequality. Simple meaning: Taking log helps in making the calculations simpler and know the behaviour of averages.

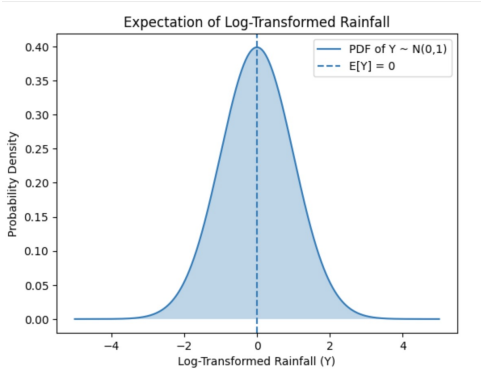
Jensen's Inequality:

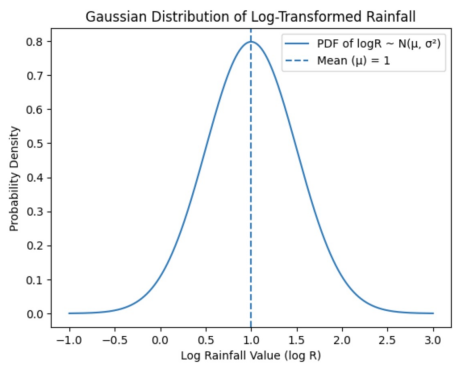
$$\mathbb{E}[e^Y] \geq e^{\mathbb{E}[Y]}$$



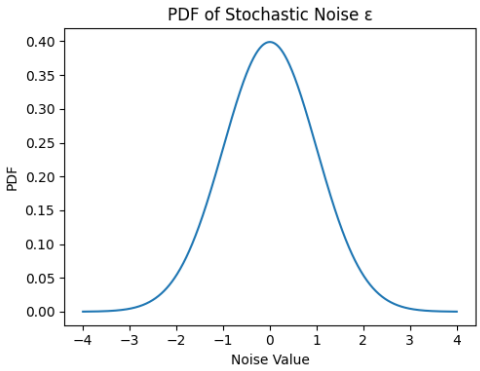


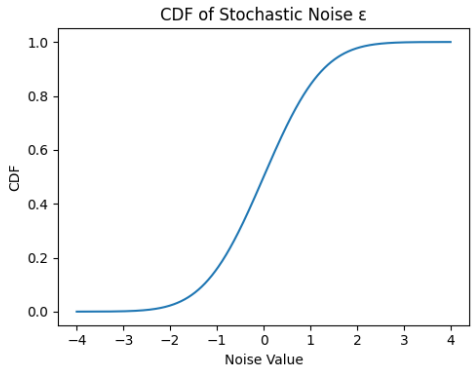


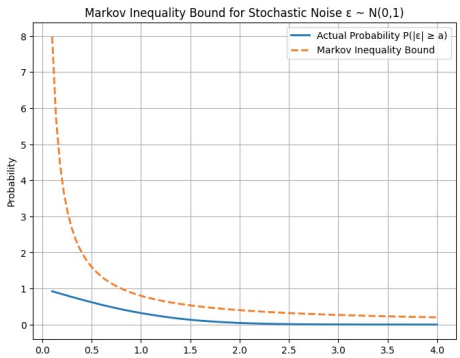


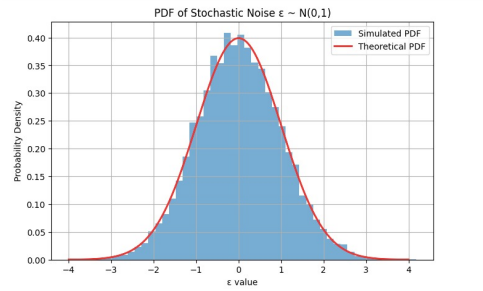


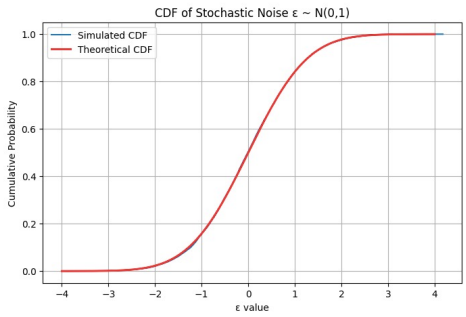
Stochastic Noise: This is what small random variations in rainfall represent. It is modelled as a zero mean normal distribution. To have a bound on large deviation due to noise, we take Markov's inequality. Simple meaning: There are always little random changes in rainfall and we make some estimates of how big they could be.











Exceedance Indicator: This variable is used for determining if rainfall exceeds or falls below a certain level:

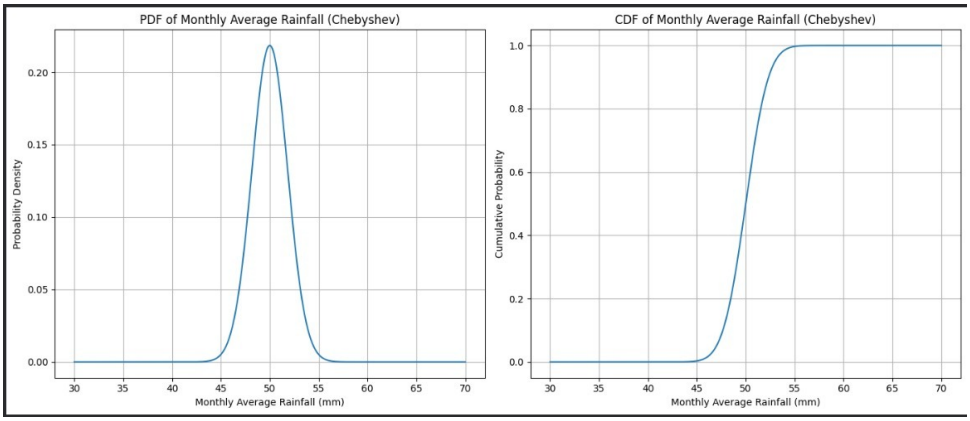
$$I = 1 \text{ if rainfall is greater than k}$$

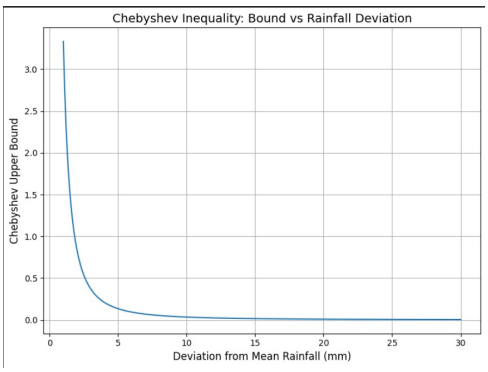
$$I = 0 \text{ if rainfall is less than k}$$

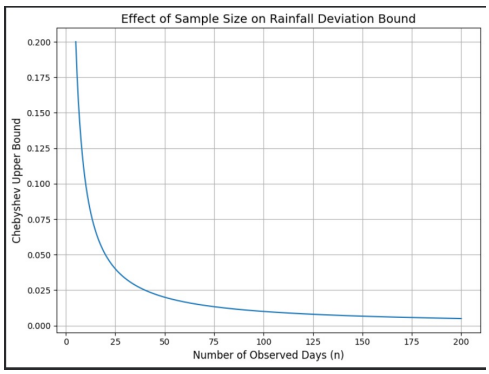
It is distributed according to a Bernoulli distribution. We use Chebyshev's inequality to obtain a bound on the uncertainty of estimating the probability of exceeding the threshold. Simple meaning: This variable simply informs us about whether or not heavy rainfall occurred and we are calculating the level of uncertainty of this probability.

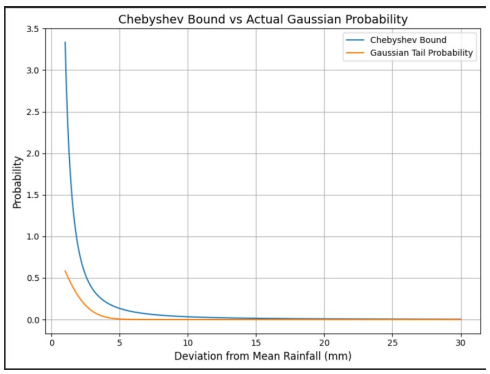
Chebyshev's Inequality:

$$P(|R - \mathbb{E}[R]| \geq k) \leq \frac{\mathrm{Var}(R)}{k^2}$$









Scribe Question 3: Probabilistic Reasoning and Dependencies

Explain how probabilistic relationships (e.g., dependence, independence, conditionality) are used within your project system to support reasoning, inference, or decision-making.

Answer:

Dependence Structure: The amount of rainfall is determined by storm motion. Spatial rainfall uncertainty is affected by the motion uncertainty. The probability of extreme rainfall is related to the intensity of rainfall and the choice of threshold.

Conditional Reasoning: The system makes logical decisions based on the probabilities that are of the type: Probability of an event (rainfall) that is over some threshold (using existing radar data). Inequality is employed to give conservative probabilistic guarantees although the actual distribution is not known.

Inequality as part of the Decision-Making process:

Each method approaches uncertainty thinking differently:

- Motion uncertainty with heavy tails is represented by Laplace distribution.
- Markov's inequality is one which uses the mean to limit the extreme rainfall risk.
- The inequality of Jensen describes the expectation in the non-linear change.
- Inequality of Chebyshev bounds deviation of exceedance.

These tools enable the forecasting of uncertainty without having precise information on the distribution.

Scribe Question 4: Model–Implementation Alignment

Describe how the current probabilistic model aligns with the project's implementation or experimental setup. Highlight any assumptions that influence design or evaluation choices.

Answer:

The probabilistic model of our project is linked to the implementation and experimental design, as both attempt to model the rainfall prediction problem as a stochastic process and not as a deterministic process. The implementation uses probabilistic distributions and inequality based bounds to estimate rainfall uncertainty and extreme rainfall risk, which directly presented the assumptions of the probabilistic model.

- The rainfall intensity is modeled to be log-normally distributed, which is in line with the experimental observation that the rainfall amount is always positive and can have extreme variability.
- The uncertainty in the storm motion is modeled to be Laplace distributed, which can handle more deviations than a normal distribution. This is in line with the experimental design, where the storm trajectory predictions have to handle sudden and unpredictable changes.
- The stochastic noise is modeled using a zero-mean normal distribution and is practical as it returns real-world variability and measurement uncertainty.
- The exceedance indicator variable that is modeled using a Bernoulli random variable is realistic and reflects the objective of the system to detect rainfall above a certain critical level.

The inequalities of Markov's and Chebyshev's, are directly applied in the implementation to give a probabilistic bound of rainfall exceedance and uncertainty. These approaches can be applied when there is incomplete information about the distribution. Lastly, the assumptions made in the probabilistic model directly affect the implementation of the system to ensure that uncertainty, variability, and extreme events are realistically modeled in the implementation.

Scribe Question 5: Cross-Milestone Consistency and Change

Describe the current state of your project's probabilistic model. Identify the assumptions, components, or relationships that are presently well-defined, and indicate which aspects

are expected to evolve or require refinement in subsequent milestones. Briefly justify your reasoning.

Answer:

In this Milestone 2, the probabilistic model has a number of well-defined random variables that serve as the basis for the final mathematical model of the rainfall prediction system. The major random variables are rainfall intensity, the uncertainty of storm motion, stochastic noise, and the exceedance indicator. The relationship between rainfall intensity and threshold exceedance probability has been appropriately defined. The application of probabilistic inequalities - Markov's inequality, Jensen's inequality, and Chebyshev's inequality, ensures theoretical validation and estimates for the uncertainty estimation in the model.

The relationship between storm motion and rainfall intensity has also been appropriately defined from the conceptual standpoint, showing that the uncertainty of storm motion has a direct impact on the rainfall distribution. This aligns with the purpose of the system to model rainfall as a function of uncertain atmospheric dynamics.

However, some aspects of the probabilistic model will change over the next milestones. For example, the actual values of the parameters of the distributions, such as the mean and variance of the intensity and motion of rainfall, may need to be adjusted based on actual data. Furthermore, the correlations between the variables may need to be adjusted to reflect more complex relationships, such as time series relationships or correlations between rainfall measurements at different locations.

The calculation of the exceedance probabilities may also need to be adjusted to incorporate more sophisticated probabilistic models or data-driven methods. These adjustments will ensure that the probabilistic model is more accurate and realistic while still being consistent with the current framework developed in this milestone 2.

Scribe Question 6: Open Issues and Responsibility Attribution

Identify any unresolved probabilistic questions or ambiguities in the project at this stage, and indicate which role(s) or tasks are responsible for addressing them in the next milestone.

Answer:

There are still some probabilistic problems and uncertainties that have not been solved yet and will be addressed in the future milestones. The most important ones are:

- **Distribution Parameter Estimation:** The parameters of the probabilistic distributions, such as the scale parameter of the Laplace distribution for the storm motion and the mean and variance of the log-normal rainfall intensity distribution, have not been estimated yet. These parameters will have to be estimated and validated using real rainfall data or simulated data to ensure the correctness of the model.

- **Dependence Modeling Between Variables:** Although it has been proved that storm motion is a contributing factor to rainfall intensity, the exact mathematical expression of the relationship between these two variables has not been completely defined yet. Future work may be needed to define conditional probability models or joint distributions to better model the dependence between variables.

- **Tightness and Accuracy of Inequality Bounds:** The above application of Markov's and Chebyshev's inequalities has provided a safe, but possibly not tight enough, probabilistic bound.
- **Consequences of Stochastic Noise and Error of Measurement:** The current model considers stochastic noise that follows a normal distribution, but the impact of noise on rainfall forecasting and the probability of exceedance will be addressed in future research to ensure that the uncertainty is captured correctly.

The challenge of addressing these issues will be allocated to various roles and tasks in the project:

- **Probabilistic Modeling Team:** The team will be allocated the responsibility of addressing assumptions about distributions, parameter estimation, and mathematical modeling of uncertainty and dependence.
- **Data Analysis Team:** The team will be allocated the responsibility of analyzing and validating rainfall and storm data to calibrate parameters and enhance the accuracy of probabilistic forecasts.
- **Implementation and Simulation Team:** The team will be allocated the responsibility of implementing enhanced probabilistic models in the system, simulating the system, and testing the system performance under uncertainty.
- **Evaluation and Validation Team:** The team will be allocated the responsibility of evaluating probabilistic forecasts, validating the boundaries of uncertainty, and ensuring consistency between the theoretical model and the implementation.